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SUMMARY

- Aristotle's idea of motion: To maintain motion, a force is required
- Galileo's idea of motion: To maintain motion, a force is not required
- Mass is a measure of inertia of the body
- Newton's first law states that under no external force, the object continues its state of motion
- Newton's second law states that to change the momentum of the body, external force is required
  Mathematically it is defined as $\vec{F} = \frac{d\vec{p}}{dt}$
- Both Newton's first and second laws are valid only in inertial frames
- Inertial frame is the one in which if there is no force on the object, the object moves at constant velocity.
- Newton's third law states that for every force there is an equivalent and opposite force and such a pair of forces is called action and reaction pair.
- To draw a free body diagram for an object,
  - Isolate the object from other objects and identify the forces acting on it
  - The force exerted by that object should not be taken into account
  - Draw the direction of each force with relative magnitude
  - Apply Newton's second law in each direction
- If no net external force acts on a collection of particles (system), then the total momentum of the collection of particles (system) is a constant vector.
- Internal forces acting in the system cannot change the total momentum of the system.
- Lami's theorem states that if an object is in equilibrium under the concurrent forces, then the ratio of each force with the sine of corresponding opposite angle is same.
- An impulse acting on a body is equal to the change in momentum of the body. Whenever a force acts on the object for a very short time, it is difficult to calculate the force. But impulse can be calculated.
- Static friction is the force which always opposes the movement of the object from rest. It can take values from zero to $\mu_s N$. If an external force is greater than $\mu_s N$ then object begins to move.
- If the object begins to move, kinetic friction comes into effect. To move an object with constant velocity, the external force must be applied to overcome the kinetic friction. The kinetic friction is $\mu_k N$.
- Rolling friction is much smaller than static and kinetic friction. This is the reason that to move an object roller coaster is fixed in the bottom of the object. Example: Rolling suitcase